Both inverse types of a map with a quasi-inverse are contractible

The Univalent Foundations Program

hott-book:equivalences.tex:local-snapshot-20261011; CC BY-SA 3.0 Unported

Editorial note. The counted statement is the author lem environment at source lines 319-321 and its proof environment opens on line 322, the next line; the pairing is by adjacency and the retained proof range 322-328 is the whole environment.

Editorial note. The retained counter values (chapter 4, section 2, theorem offset 8) make the counted lemma print as Lemma 4.2.9, matching the original book numbering read from the label-number dump of the pinned snapshot (newlabel lem:inv-hprop = 4.2.9).

Editorial note. Three labels the proof cites lie outside the card and are declared in external_references as non-load-bearing navigation pointers: thm:equiv-compose-equiv (book Lemma 4.2.8, defined at line 308 of this chapter pinned file), thm:equiv-iso-adj (book Theorem 4.2.3, defined at line 215 of the same file) and thm:contr-hae (book Theorem 4.2.6, defined at line 265 of the same file).

Editorial note. The retained preamble is the book own macro library and layout adapter as stored for this chapter; no package outside it is requested.

Editorial scope. One original result of the pinned HoTT Book source tree, counted once: contractibility of the two inverse types of a map with a quasi-inverse (author label lem:inv-hprop, book Lemma 4.2.9).

Allowed standard prerequisites (editorial).

- Left and right inverse types. The type of left inverses of f is the type of maps g with g composed with f equal to the identity, and the type of right inverses is the same with the composite in the other order.. They are the two types shown contractible.
- Fibre of a precomposition map. The fibre of the map sending h to h composed with f over the identity is exactly the type of left inverses of f, up to the equivalence supplied by function extensionality.. It identifies the inverse type with a fibre.
- Fibres of equivalences are contractible. The fibre of a map that is an equivalence over any point is contractible.. It finishes each of the two cases.

Source reference. Original lemma 4.2.8: sources/hott-equivalences/pinned-original-source.tex, source lines 308–308. Book Lemma 4.2.8, that a composite of equivalences is an equivalence, cited at source line 325. Navigation pointer only: the retained proof is complete and the step it supplies is one line of it.

Source reference. Original theorem 4.2.3: sources/hott-equivalences/pinned-original-source.tex, source lines 215–215. Book Theorem 4.2.3, relating the several notions of equivalence, cited at source line 325. Navigation pointer only.

Source reference. Original theorem 4.2.6: sources/hott-equivalences/pinned-original-source.tex, source lines 265–265. Book Theorem 4.2.6, contractibility of the half adjoint equivalence structure, cited at source line 325. Navigation pointer only.

Lemma 4.2.9. *If $f : A \to B$ has a quasi-inverse, then the types $\mathsf{rinv}(f)$ and $\mathsf{linv}(f)$ are contractible.*

Proof. By function extensionality, we have

$$\mathsf{linv}(f) \simeq \sum_{g:B\to A} (g \circ f = \mathsf{id}_A).$$

But this is the fiber of $(-\circ f)$ over id_A, and so by original lemma 4.2.8 and original theorems 4.2.3 and 4.2.6, it is contractible. Similarly, $\mathsf{rinv}(f)$ is equivalent to the fiber of $(f \circ -)$ over id_B and hence contractible. □